

Approximating the Score Function using Semi-Discrete Optimal Transport

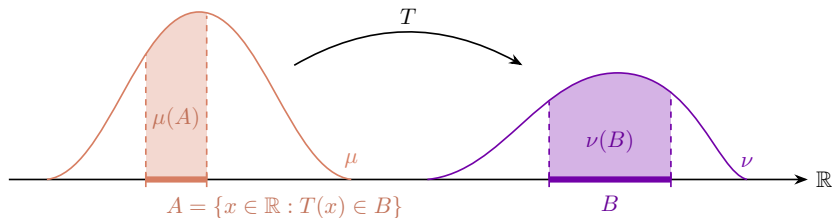
Curves and Surfaces 2026

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Laboratoire de Mathématiques d'Orsay

June 11th

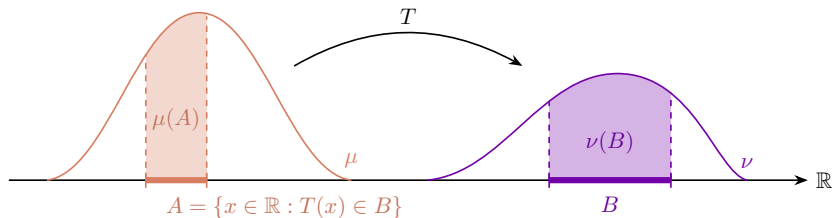
Quick reminders on Optimal Transport 1/2



- ▶ Given ground cost $c(x, y) = \|x - y\|^2$
- ▶ Among all maps T satisfying $\forall B, \nu(B) = \mu(T^{-1}(B))$
- ▶ Find the one minimising the average transport cost

$$W_2(\mu, \nu)^2 := \inf_{T: T_{\#}\mu = \nu} \int_{\mathbb{R}^d} \|x - T(x)\|^2 \mu(dx)$$

Quick reminders on Optimal Transport 2/2

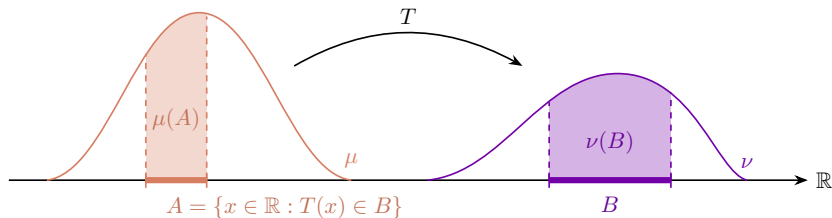


► Kantorovich duality

$$\inf_{T: T_{\#}\mu=\nu} \int_{\mathbb{R}^d} \|x - y\|^2 \mu(dx) = \sup_{\phi} \int \phi(x) \mu(dx) + \int \phi^c(y) \nu(dy),$$

where $\phi^c(y) := \inf_x c(x, y) - \phi(x)$.

Quick reminders on Optimal Transport 2/2



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► Let $c(x, y) = \alpha \|x - y\|^2$

$$\nabla_x c(x, T^*(x)) = \nabla \phi^*(x) \iff T^*(x) - x = \boxed{y - x = -\frac{1}{2\alpha} \nabla \phi^*(x)}$$

Displacement

What is the score function?

- **Forward process:** Ornstein–Uhlenbeck (OU) process:

$$dX_t = -X_t dt + \sqrt{2} d\omega_t, \quad X_0 \sim \mu, \quad t \in [0, T].$$

Let $\rho_t := \text{Law}(X_t)$ and $\gamma \propto e^{-\|\cdot\|^2/2}$. Then ρ_t solves the Fokker–Planck equation

$$\partial_t \rho_t(x) = \Delta \rho_t(x) + \text{div}(\rho_t(x)x). \quad (\star)$$

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- **Macroscopic POV:** Equation $(\star)$ is the 2-Wasserstein Gradient Flow of the relative entropy $\mathcal{H}(\rho|\gamma) := \int \frac{\|x\|^2}{2} \rho + \int \rho \ln(\rho)$. It means $(\star)$ can be written as

$$\partial_t \rho_t(x) = -\nabla \cdot (\rho_t(x) v_t(x)),$$

where $v_t(x) = -x - \nabla \ln(\rho_t(x))$.

Using OT to learn the score 1/2

- ▶ Given a time-step $\tau > 0$, such a gradient flow can be discretised in time using the **JKO scheme** (or Implicit Euler scheme, or Proximal step)

$$\begin{cases} \rho_0 = \mu \\ \rho_{n+1}^\tau := \arg \min \mathcal{H}(\rho|\gamma) + \frac{1}{2\tau} W_2(\rho_n^\tau, \rho)^2 \end{cases}$$

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- ▶ **At step n :** We show that

$$\rho_{n+1}^\tau \propto \exp\left(-\phi^{c/2\tau}\right) d\gamma$$

where ϕ is such that

$$\inf_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} d\gamma\right) - \int \phi d\rho_n^\tau.$$

Using OT to learn the score 2/2

Recall that:

$$\rho_{n+1}^\tau \propto \exp\left(-(\phi^\star)^{c/2\tau}\right) d\gamma \quad \text{where} \quad \phi^\star \in \arg \min_{\phi} \tilde{J}(\phi) := \ln\left(\int e^{-\phi^{c/2\tau}} d\gamma\right) - \int \phi d\rho_n^\tau$$

- The optimal potential $\phi^\star$ satisfies ($\alpha = \frac{1}{2\tau}$ here)

$-\tau \nabla \phi^\star(x) = y - x$
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Displacement

- The displacement of the particle x being $\tau v_t(x)$, we obtain

$$-\tau \nabla \phi^\star(x) = \tau v_t(x) = -\tau(x + \nabla \ln(\rho_{n+1}^\tau(x))) \iff \boxed{\nabla \ln(\rho_{n+1}^\tau(x)) = \nabla \phi^\star(x) - x}.$$

General contributions

Proposition (G.)

The functional $\tilde{J}(\phi) := \ln \left(\int e^{-\phi^{c/2\tau}} d\gamma \right) - \int \phi d\rho_n^\tau$ is convex.

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Theorem (G.)

Let

$$J(\phi) := \int e^{-\phi^{c/2\tau}} d\gamma - \int \phi d\rho_n^\tau.$$

Then,

$$\arg \min_{\phi} J(\phi) \subseteq \arg \min_{\phi} \tilde{J}(\phi) \quad \text{and} \quad \arg \min_{\int e^{-\phi^{c/2\tau}} d\gamma = 1} \tilde{J}(\phi) = \arg \min_{\phi} J(\phi).$$

Stochastic gradient descent 1/2

$$\blacktriangleright J(\phi) = \mathbb{E}_{\mathbf{y} \sim \gamma} \left[\exp \left(-\phi^{c/2\tau}(\mathbf{y}) \right) - \int \phi(\mathbf{x}) \rho_n^\tau(\mathrm{d}\mathbf{x}) \right]$$

Stochastic gradient descent 1/2

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- ▶ Choice of parametrisation: semi-discrete optimal transport using

$$\hat{\rho}_n^\tau = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{x}_i^{(n)}} \rightsquigarrow \begin{cases} \phi(x; \theta) = \theta^\top \Phi(x) \\ \Phi(x) = (\Phi_1(x), \dots, \Phi_N(x)) \\ \Phi_i(x) = \mathbb{1}_{\{x = \hat{x}_i^{(n)}\}} \end{cases}$$

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► $G(\theta) = \mathbb{E}_{y \sim \gamma} \left[\exp \left(- \min_{1 \leq i \leq N} \left\{ \frac{c(\hat{x}_i^{(n)}, y)}{2\tau} - \theta_i \right\} \right) - \frac{1}{N} \sum_{i=1}^N \theta_i \right]$

Stochastic gradient descent 2/2

Projected SGD algorithm ($c_\tau(\mathbf{x}, \mathbf{y}) := \frac{c(\mathbf{x}, \mathbf{y})}{2\tau}$)

$$\left\{ \begin{array}{l} \text{pick } \mathbf{y} \sim \gamma \\ \text{let } i := \arg \min_j c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y}) - \theta_j \\ \text{define } \mathbf{e}_i \text{ as the } i\text{-th canonical basis vector of } \mathbb{R}^N \\ \theta^{(k+1)} := \text{proj}_{B(0, R_{\text{proj}})} \left(\theta^{(k)} - \alpha_k e^{\theta^{(k)} - c_\tau(\hat{\mathbf{x}}_j^{(n)}, \mathbf{y})} \mathbf{e}_i + \frac{\alpha_k}{N} \mathbb{1} \right) \end{array} \right.$$

Theorem (G .)

$$\triangleright \alpha_k = \frac{C}{\sqrt{k}}, \quad \hat{\theta}_K := \frac{1}{K} \sum_{k=0}^K \theta^{(k)}$$

Then, it holds that

$$0 \leq \mathbb{E}[G(\hat{\theta}_K)] - G(\theta^*) \leq \frac{2R_{\text{proj}}^2}{CK} + \frac{B^2C}{\sqrt{K}} = \mathcal{O}(K^{-\frac{1}{2}}),$$

where θ^* denotes the unique minimiser of G .

Approximating the next JKO iterate

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$$\hat{\rho}_n = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{x}_i^{(n)}} \xrightarrow{\text{JKO}_\tau} \tilde{\rho}_{n+1}^\tau(dy) \propto \max_i e^{\theta_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(dy)$$

Approximating the next JKO iterate

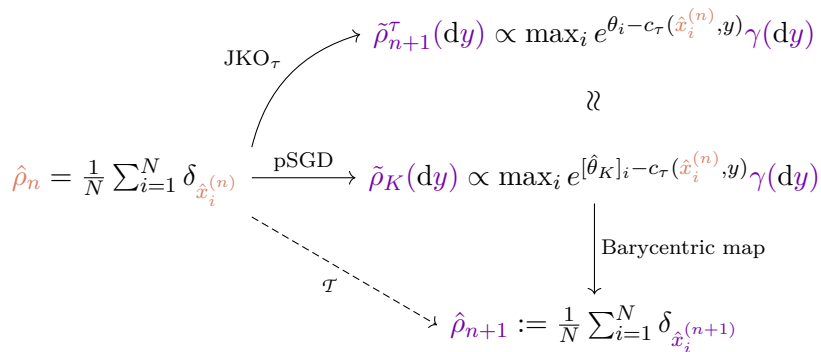
$$\begin{array}{ccc} & \text{JKO}_\tau & \\ & \nearrow & \\ \hat{\rho}_n = \frac{1}{N} \sum_{i=1}^N \delta_{\hat{x}_i^{(n)}} & \xrightarrow{\text{pSGD}} & \tilde{\rho}_K(\mathrm{d}y) \propto \max_i e^{[\hat{\theta}_K]_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y) \\ & & \Downarrow \\ & & \tilde{\rho}_{n+1}^\tau(\mathrm{d}y) \propto \max_i e^{\theta_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y) \end{array}$$

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 & & \Downarrow \text{Barycentric map} \\
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 \end{array}$$

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 $\Re$

Approximating the next JKO iterate



More on the barycentric map

- $\tilde{\rho}_K(\mathrm{d}y) \propto \max_i e^{[\hat{\theta}_K]_i - c_\tau(\hat{x}_i^{(n)}, y)} \gamma(\mathrm{d}y)$ is **piecewise log-quadratic**

More on the barycentric map

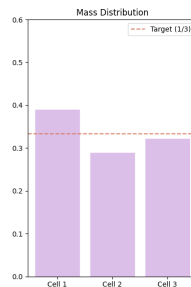
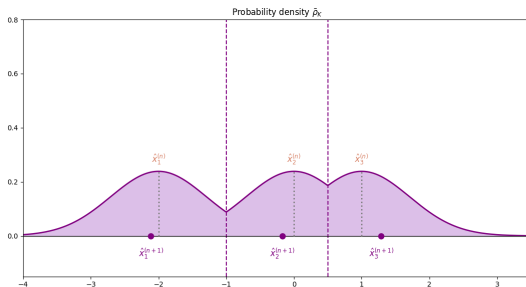
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- ▶ **Laguerre cells:**
$$L_i(\theta) := \left\{ y \in \mathbb{R}^d : \forall 1 \leq i \leq N, c_\tau(\hat{x}_i^{(n)}, y) + \theta_i \leq c_\tau(\hat{x}_j^{(n)}, y) + \theta_j \right\}$$

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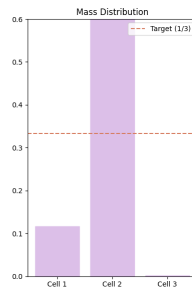
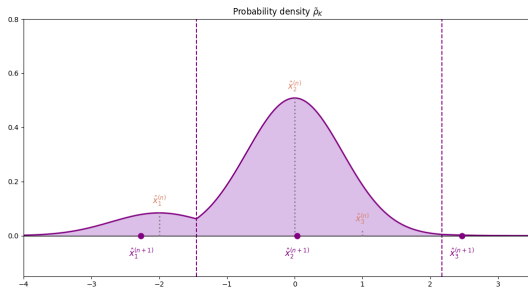


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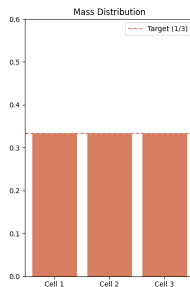
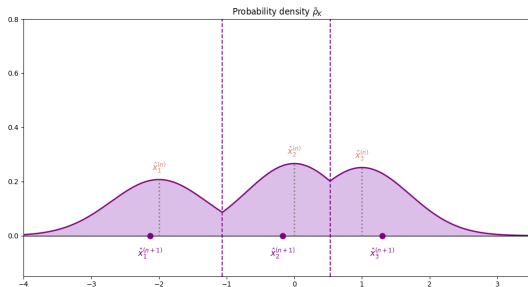


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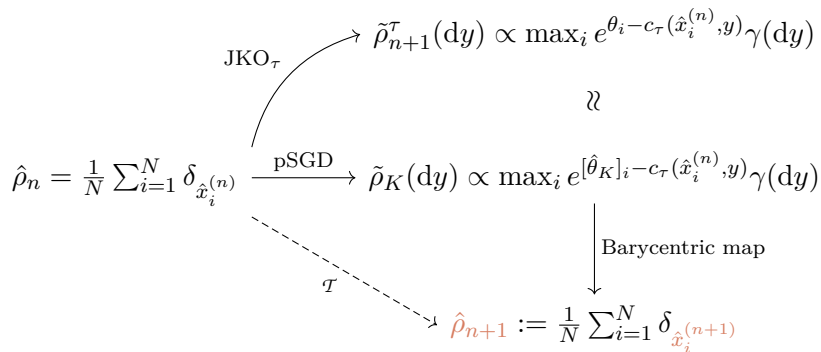
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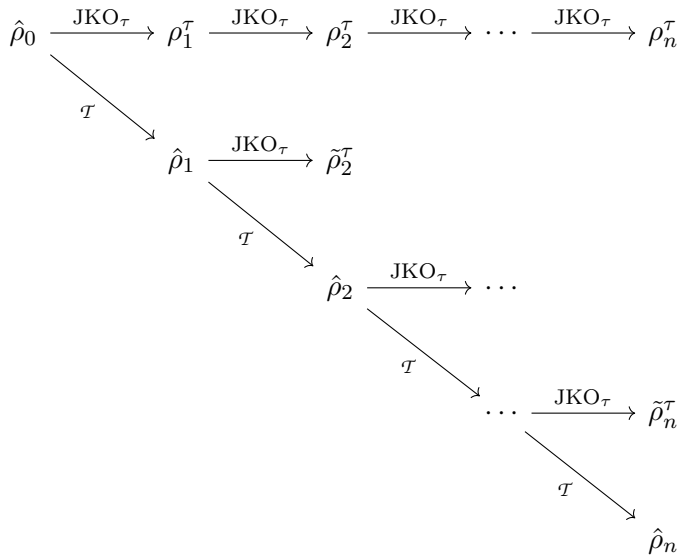
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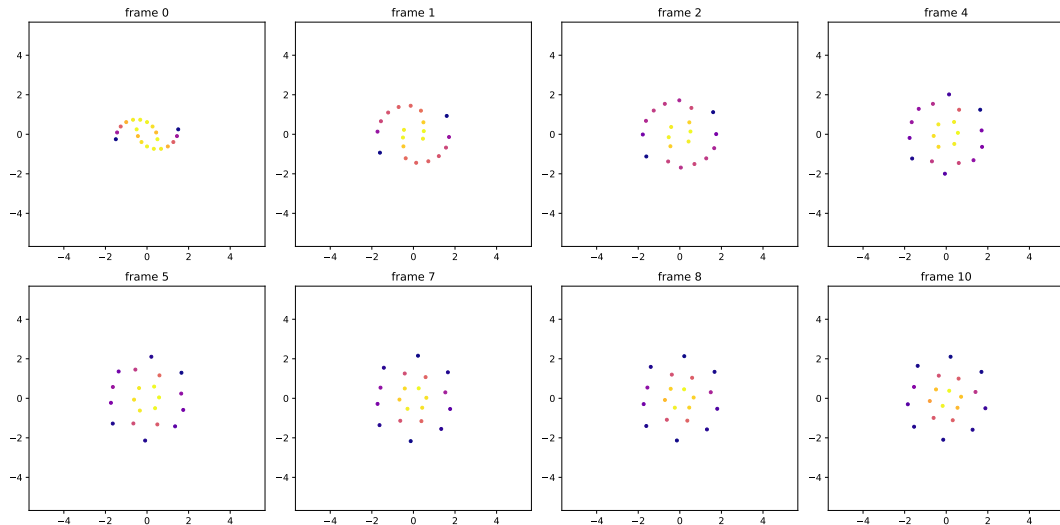
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Iterating the whole procedure



Some “animations”



How to reverse the flow???

- ▶ The true score is piecewise linear

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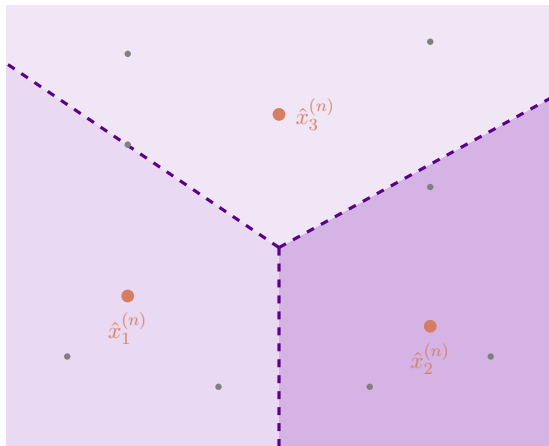
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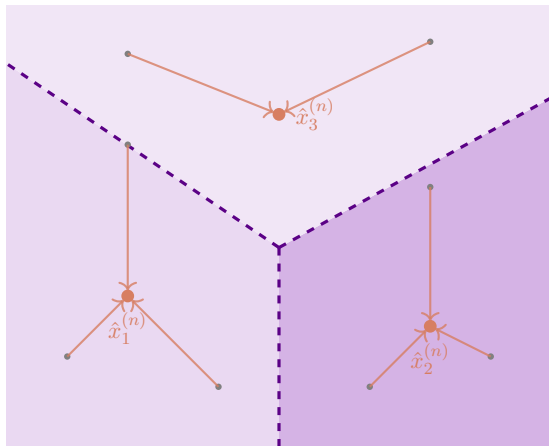
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Thank you for your attention!